

MAGNETOM

Replacement of Parts System

Cooling

Harmony syngo MR
Symphony syngo MR
Sonata syngo MR
Trio
Upgrades to syngo MR:
from Impact/Expert
from Vision

Document Version

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1.1 Replacement of the Cooler

This chapter describes the repair procedure for the cooler in the RCA and the cabinet cooling system (CCS).



Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" and the protective earth wire connected to the "main component".

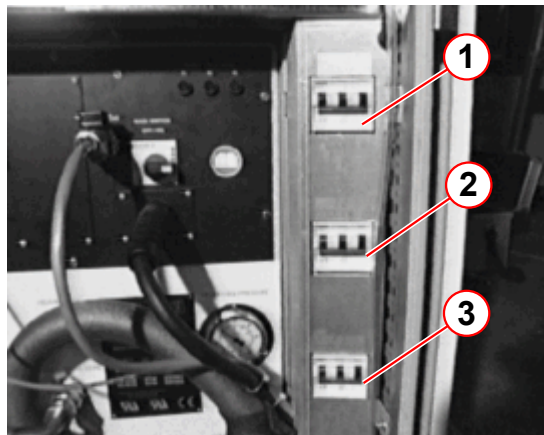
Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

1.1.1 Auxiliary tools

- Spanner: 30 mm; 10 mm
- Screwdrivers
- Adjustable spanner
- Pipe wrench, adjustable
- Bucket

1.1.2 Removing the Cooler

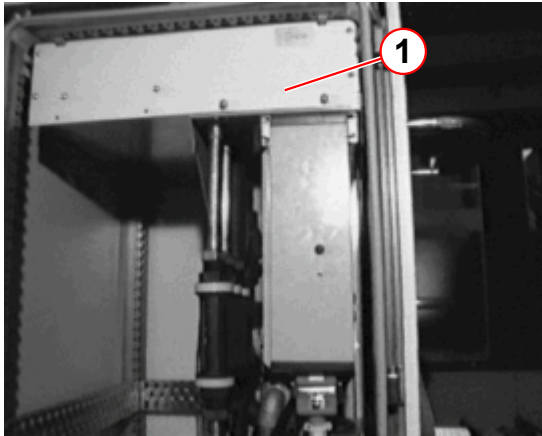
Fig. 1: Circuit breaker



- (1) F1
- (2) F2
- (3) F3

- Switch off the helium compressor
- Switch off F22 in CCA.
- Switch off the voltage supply to the Cooler via circuit breaker F1/F2 "Cooler 1" / "Cooler 2" in the RCA.
- Switch off F4 in the mainsbox on top off the CCA.
- Turn off the primary water supply

Fig. 2: Front cover



(1) Front cover

- Remove the front cover.
- Remove the top cover.

- Drain the system refer to: (→ Drain procedure of the cooler / Page 61).

NOTICE

Cooling water containing citric acid should be collected while draining the Cooler.

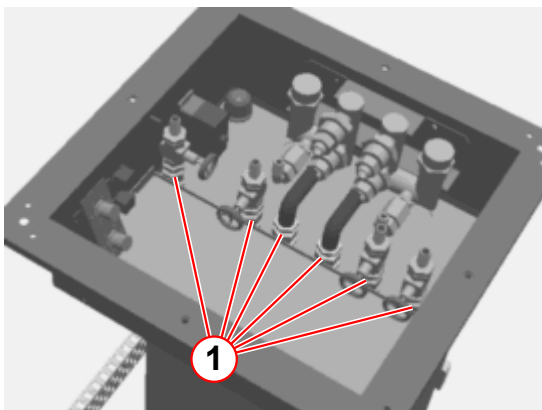
- » This water should be refilled into the secondary cooling circuit after new cooler is installed.

NOTICE

Disposing the cooling water.

- » Follow the country-specific guidelines when disposing of cooling water containing citric acid

Fig. 3: Water distribution top of RCA



(1) Union end

- Disconnect the water distribution from the cooler.
Disconnect the union ends.
- Disconnect cable SK1 and SK2.
- Remove the 2 screws in the base plate for the cooler.
- Remove the cooler from the front of the RCA.



Cooler weight approx. 100 kg

1.1.3 Installation of the new cooler

- Install the new cooler
- Attach the cooler to base plate

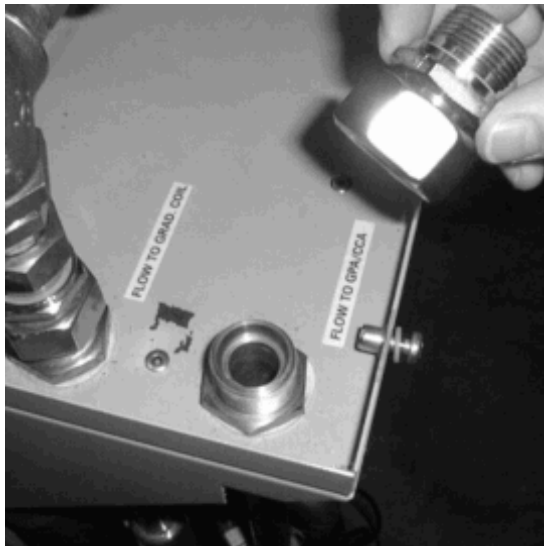
NOTICE

If an old type of cooler is replaced for a new type of cooler.

» An adapter kit is necessary because the type of couplings are different

- If necessary, installing the adapter kit

Fig. 4: Adapter



- Place the adapter to
 - Flow to GPA/CCA
 - Flow to Grad. Coil
 - From GPA/CCA
 - From Grad. Coil
- Tighten the adapter.

Fig. 5: Gasket



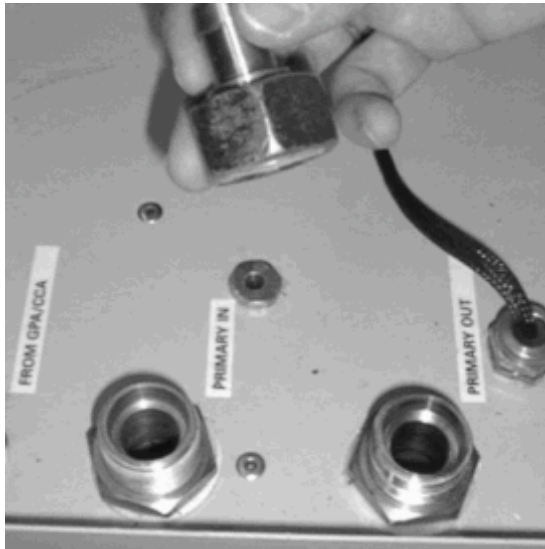
- Place the four gaskets.

Fig. 6: Valves



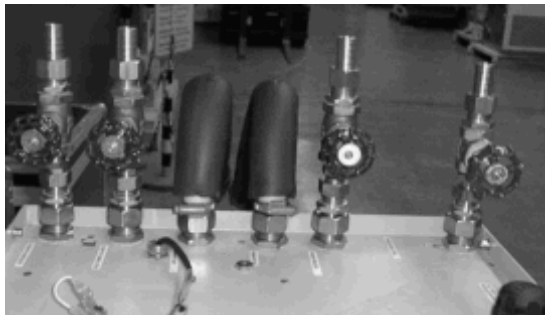
- Connect the valves with the adapter.

Fig. 7: Primary water connection



- Place the new adapter for
 - PRIMARY IN
 - PRIMARY OUT
- Reconnect the water hoses, use the clamps.

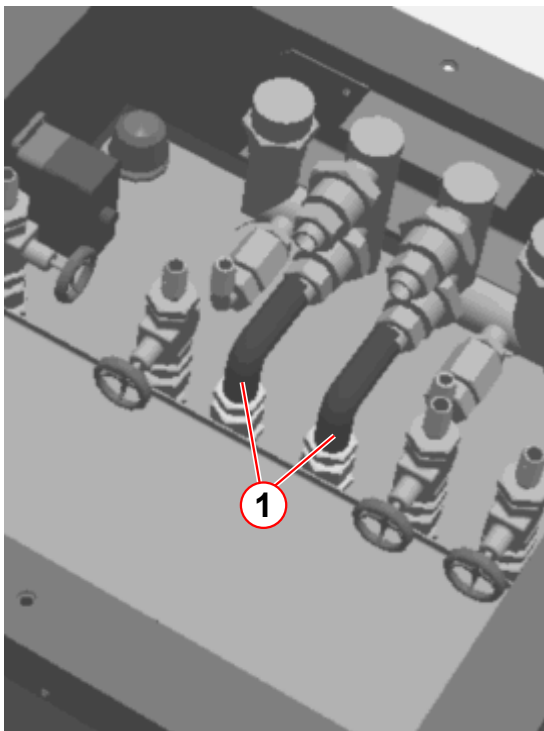
Fig. 8: Adapter kit



- Overview of the installed adapter kit.

- Reconnect the cooler to the water distribution

Fig. 9: Water hoses



(1) Water hoses

NOTICE

Tighten the water hoses.

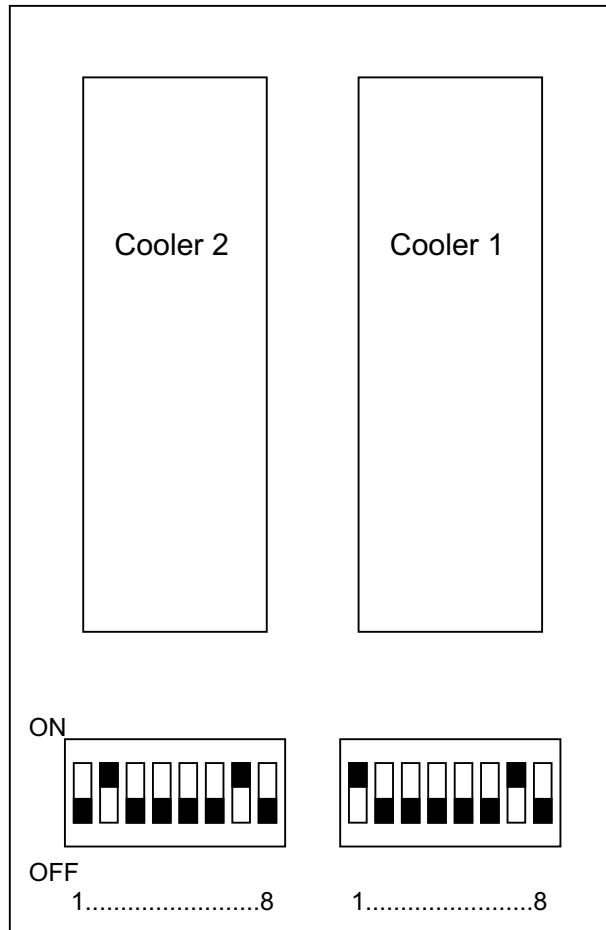
» **Don't twist the water hoses when you tighten them.**

- Connect SK1 and SK2.
- Check the DIP switch setting.
 - » Set the DIP switch in the right position.



Spare coolers are always preset as cooler 2!

Fig. 10: DIP switch setting



- Open the valves.
- Switch on the primary water supply.
- Switch on F4, F22, F1 / F2.



The controller itself will automatically determine its address by the dip switches during power up. A red blinking LED indicates normal operation of the Satchwell controller.

- Switch on the compressor.
- Start up the cooler (refer to: (→ Start-up / M1-010.815.01)).
- (→ Filling the secondary water circuit / Page 64)
- Adjust the flow rates.
- Check for water leaks



Drain water from the pipes, before returning the replaced unit



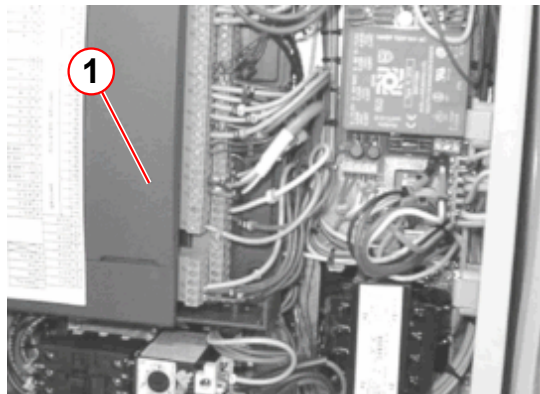
The new model water pump inside the RCA cooler switches on with a high current surge causing the circuit breaker(s) F1 and F2 (twin cooler) to trip.

Therefor the C-type circuit breaker(s) must be exchanged against a new type with a better current surge characteristics.

Every spare cooler p/n 4763566 will be delivered with two extra new D-type circuit breakers. This breaker should be fitted if the C-type breaker is still installed.

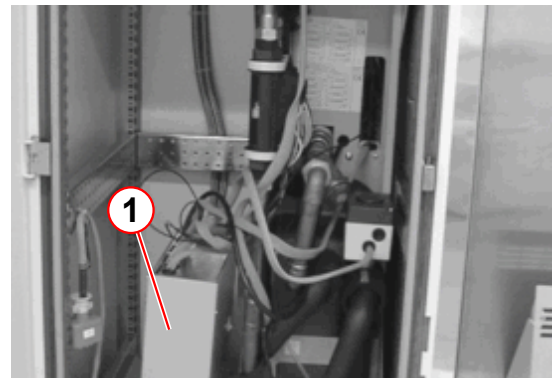
1.2 Replacement of the Level Sensor LI1

Fig. 11: Electronic box



(1) Satchwell

Fig. 12: Cooler

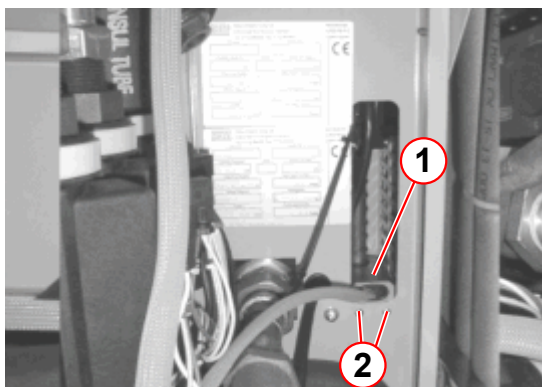


(1) Electronic box

1.2.1 Removing the level sensor

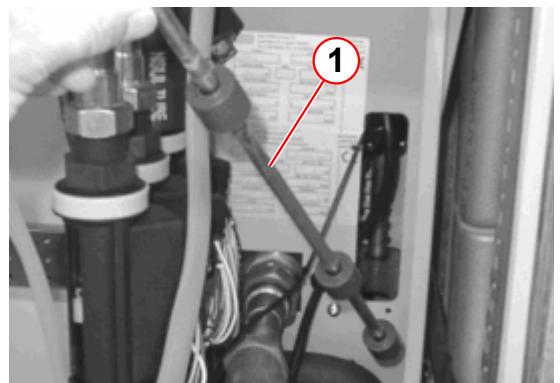
Step	Instructions	Required Tools
1	Switch off cooler Switch off F1 Switch off F22 in the CCA	
2	Disconnect sensor from the Satchwell controller PIN 17 (Yellow) and Ground (Violet) PIN 15 (Red) and Ground (Violet) PIN 14 (Brown) and Ground (Violet)	
3	Take off electronic box	

Fig. 13: Level sensor



(1) Level sensor
(2) Screws

Fig. 14: Level sensor



(1) Level sensor

Step	Instructions	Required Tools
4	Loosen two screws	Screwdriver
5	Pull the sensor from the reservoir	

1.2.2 Installing the sensor

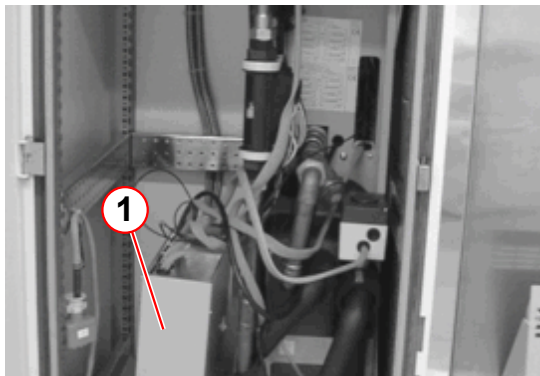
- Install the sensor in the reverse order.

1.2.3 Final checks

- Check the fill level via the SeSo.
- Add water, if necessary.

1.3 Replacement of the 3-way valve actuator M2

Fig. 15: Cooler



(1) Electronic box



You do not need to drain the water circuit

1.3.1 Removing the valve actuator

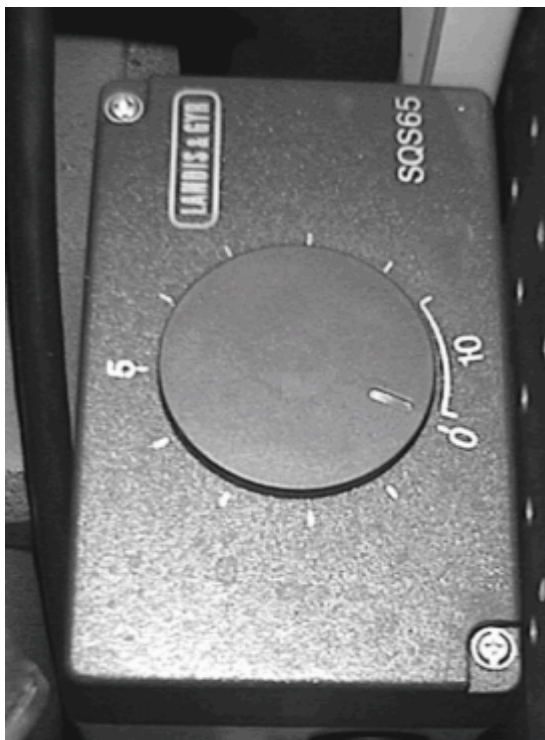
Step	Instructions	Required Tools
1	Switch off cooler Switch off F1 Switch off F22 in the CCA	
2	Remove the electronic box	
3	Disconnect cable from actuator PIN Go (Violet) PIN Y (Grey) PIN G (Red)	Screwdriver
4	Remove actuator from 3-way valve	Spanner 32 mm

1.3.2 Installing the valve actuator

- Install in the reverse order.

1.3.3 Final check

Fig. 16: Actuator



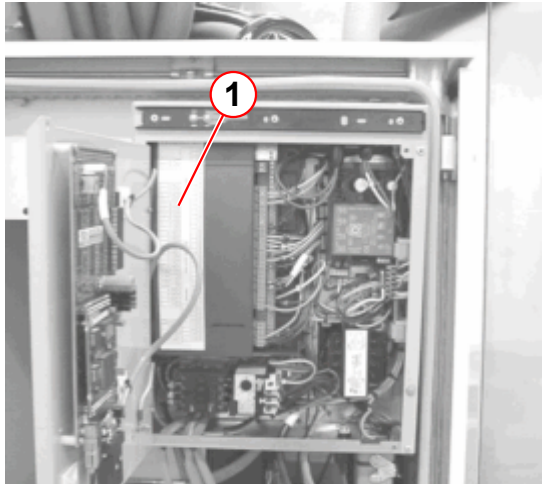
Start the cooler and test the function of the actuator

- Check the operating of the actuator.
"0" : heat exchanger is bypassed
"10": bypass is closed.

1.4 Replacing the Temperature sensor TI1 or TI2

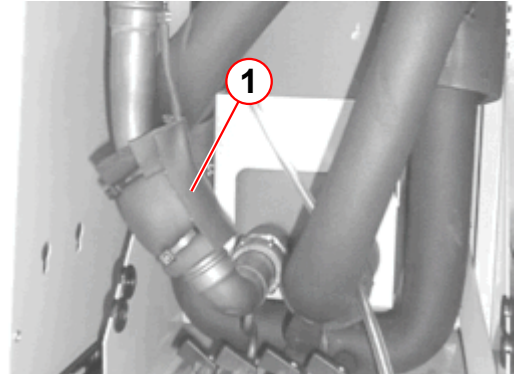
1.4.1 Removing the temperature sensor

Fig. 17: Electronic box



(1) Satchwell

Fig. 18: Temperature sensor



(1) Temperature Sensor of the secondary circuit

Step	Instructions
1	Switch off cooler Switch off F1 Switch off F22 in the CCA
2	Disconnect sensor at the Satchwell controller TI2 for secondary circuit; PIN 22 and Ground TI1 for primary circuit; PIN 23 and ground (TI1 is located under the water filter of the primary circuit)
3	Strip insulation, cut open cable ties Don't damage the insulation!

1.4.2 Installing the temperature sensor

- Install in the reverse order.



The sensor should be well insulated.
Ensure that the sensor fits snug at the coppertubing and is making good contact.

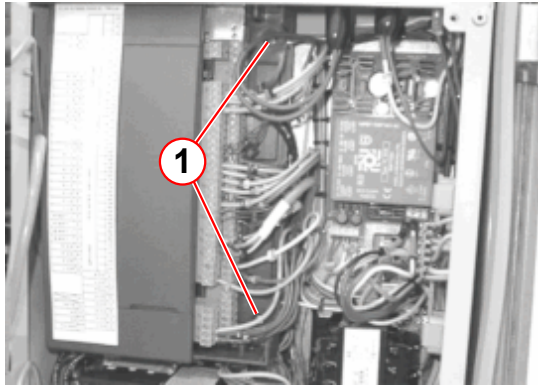
1.4.3 Final checks

- Start the cooler and test the function of the sensor.
- Check the temperature via SeSo.

1.5 Replacing the Controller (Satchwell)

1.5.1 Removing the Controller

Fig. 19: Electronic box



(1) Screw

Fig. 20: Satchwell

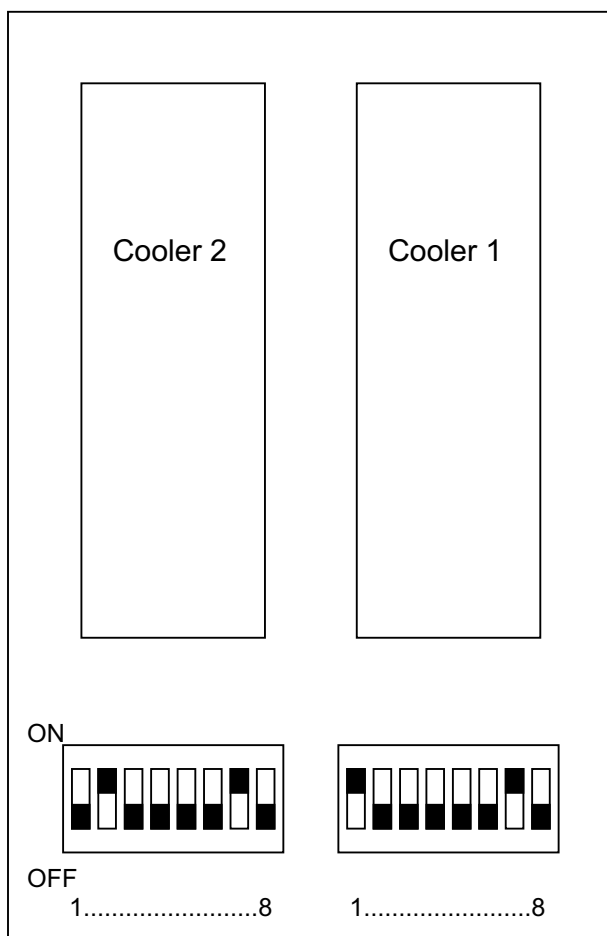


Step	Instructions	Required tools
1	Switch off cooler Switch off F1/ F2 in the RCA Switch off F22 in the CCA	
2	Remove clamps; (→ Fig. 20 Page 19)	Screwdriver
3	Remove screws; (→ Fig. 19 Page 19)	Screwdriver
4	Take out the Satchwell	

1.5.2 Installing the controller

- Install in the reverse order.
- Check the DIP switch setting.
 - » Set the DIP switch in the right position.(→ 1/ Fig. 22 Page 21)

Fig. 21: DIP switch setting



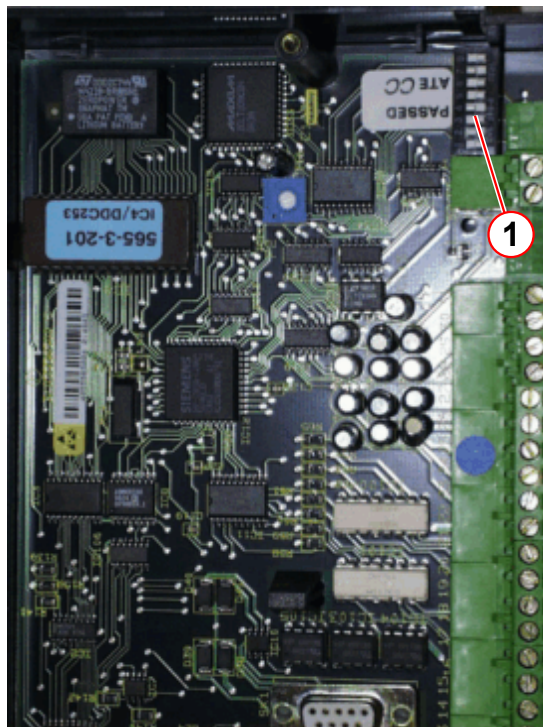
- Switch ON the 3 phase supply in the RCA, fuse F1 / F2.
- Switch on the single phase supply to Satchwell controller, fuse F22.



The controller itself will automatically determine its address by the dip switches during power up.

A red blinking LED indicates normal operation of the Satchwell controller.

Fig. 22: Satchwell controller board, DIP switches



(1) Dip switch

1.5.3 Final checks

- Start the cooler and test the function of the controller.
- Perform a "Reload all".

1.6 Replacing the flow meter



This procedure describes the replacement of the plastic flow meter with a brass flow meter

Fig. 23: Plastic Flow Meter

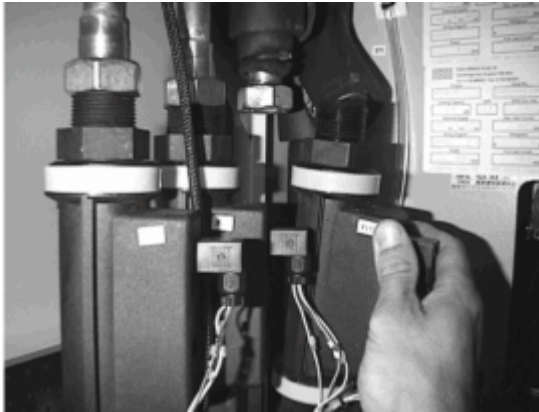
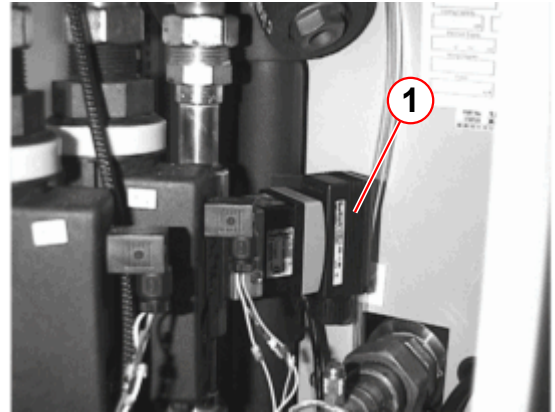


Fig. 24: Brass Flow Meter



(1) Brass Flow Meter

NOTICE

The dimensions of the brass flow meter are different.

» Please note the rules.

Rules for replacing the plastic flow meter with the brass type:

- Primary flow meter can be replaced as an individual.
- Gradient coil flow meter: primary flow meter has to be a brass type.
- CCA/GPA flow meter: primary and gradient coil flow meter has to be a brass type.

1.6.1 Required tools

- Spanner 37 mm / 42 mm
- Screwdriver

Fig. 25: Plan view of brass flow meter electrical plug Typ 8022

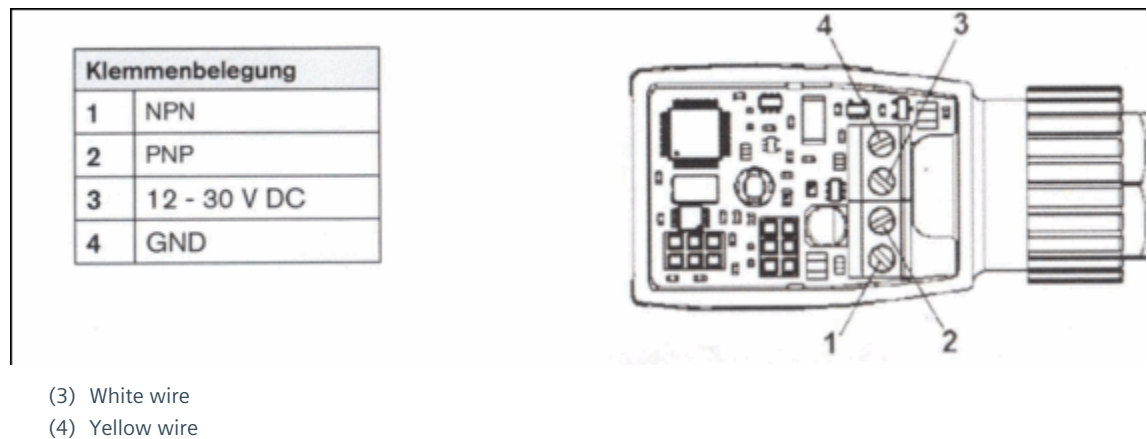
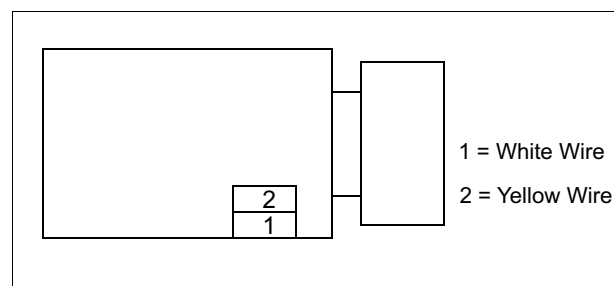


Fig. 26: Plan view of brass flow meter electrical plug Typ 8023



Step	Instructions
1	Switch off cooler Switch off F1 in the RCA Switch off F22 in the CCA
2	Replacing the primary flowmeter: Isolate the primary circuit by closing the primary water, supply and return on the Manifold (top of the RCA)
2	Replacing the secondary flowmeter: Isolate the secondary circuit by closing the Gradient coil and the CCA/GPA flow and the return.
3	Drain the cooler using drain valve. Secondary cooling water containing citric acid should be collected while draining. This water should be refilled into the secondary cooling circuit.
4	Undo the union nuts either end of the flowmeter Note: When undoing these nuts, the black plastic hexagon on the flowmeter should be retained.
5	Remove the flowmeter Discard the old sealing washers Clean the sealing faces of the pipe work. The old fibre sealing washers should be discarded from the sealing face of the pipe work

Step	Instructions
6	Remove the electrical plug from the flowmeter and diconnect the wires from the plug.
7	Connect the wires into the plug of the brass flow meter. Refer: (→ Tab. 1 Page 24); (→ Tab. 2 Page 24) or (→ Tab. 3 Page 24)
8	Install the new flow meter, flow direction can be either way. Use new washers top and bottom
9	Tighten union nuts top and bottom
10	Close the drain valves
11	Open the manifold valves
12	Switch on F1 / F22

The new wiring should be as follows:

Tab. 1 Primary Flow Meter

Wire no.	Colour	Location	Notes
28	Yellow	Pin 28	Same Location
128	Violet	Earth (pin 64)	Not required
46	White	+24V on power board	Same Location

Tab. 2 Electronic Suit Flow Meter

Wire no.	Colour	Location	Notes
30	Yellow	Pin 30	Same Location
130	Violet	Earth (pin 66)	Not required
46	White	+24V on power board	Same Location

Tab. 3 Gradient Coil Flow Meter

Wire no.	Colour	Location	Notes
29	Yellow	Pin 29	Same Location
129	Violet	Earth (pin 65)	Not required
46	White	+24V on power board	Same Location

The resitor must be kept in the position (between the terminal pin and earth). The easiest way to remove the Violet wire is to cut it back at the flow meter end.

1.7 Replacing of the Flow Meter Electronic Module



The system does not need to be drained.

Fig. 27: Flow Meter Electronic Module



Fig. 28: Removing Electronic Module

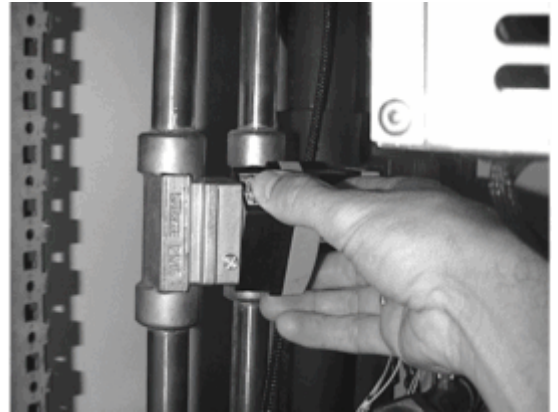


Fig. 29: Electronic Module



Step	Instructions
1	Switch off cooler. Switch off F1 in the RCA.
2	Undo the locking screw on the side of the flow meter body. (→ Fig. 27 Page 25)
3	Twist the electronic module 30 anti-clockwise and withdraw from the flow meter body. (→ Fig. 28 Page 25)
4	Disconnect the wires from the terminal block in the electronics module plug. Noting which cable colour goes to which terminal. (→ Fig. 25 Page 23), (→ Fig. 26 Page 23)

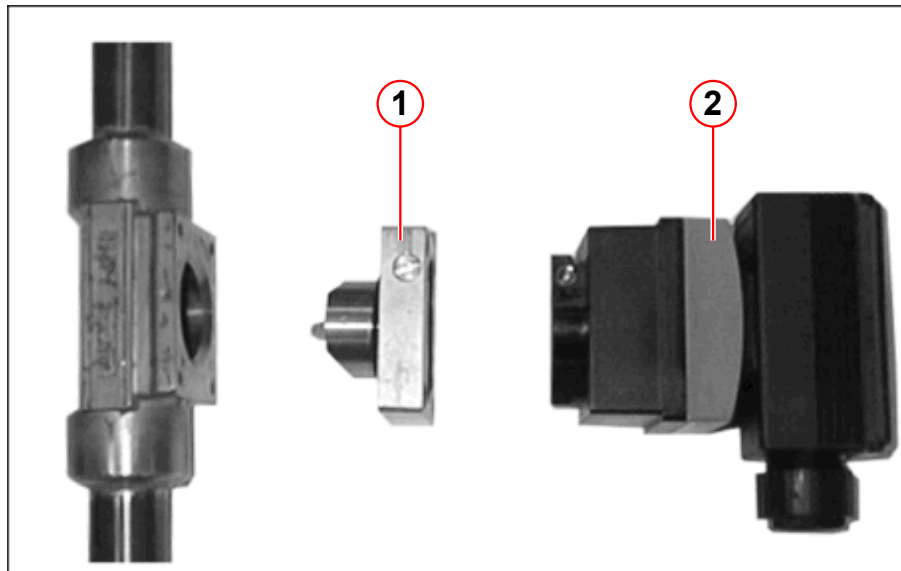
Step	Instructions
5	Connect wires to terminal block of the new electronic module plug
6	Insert electronic module back into flow meter body and twist 30° clockwise to locate
7	Tighten screw on side of flow meter to lock in position
8	Switch on F1 Switch on the cooler

1.8 Replacing the Flow Meter Paddle Wheel



The cooler must be isolated, and the water circuit fully drained before the paddle wheel is removed

Fig. 30: Parts of the Flow Meter



- (1) Paddle Wheel
- (2) Electronic Module

Step	Instructions
1	Switch off cooler Switch off F1 in the RCA
2	Replacing the primary paddle wheel: Isolate the primary circuit by closing the two valves on the manifold (top of the RCA)
2	Replacing the secondary paddle wheel: Isolate the secondary circuit by closing the gradient coil and the CCA/GPA valve and the return valve on the manifold.
3	Drain the water circuit using drain valves. Secondary cooling water containing citric acid should be collected while draining. This water should be refilled into the secondary cooling circuit.
4	Remove the electronic module.
5	Undo the four screws and withdraw the paddle wheel.
6	Check the "O"ring seat is clean in the body, ensure "O"ring on the new paddle wheel assembly is in good condition.

Step	Instructions
7	Insert the new paddle wheel assembly into flow meter body. Notice: Orientation is unimportant as flow can be in either direction. The electronic module locking screw, however, should be on the accessible side.
7	Refit the electronic module.
8	Close the drain valves.
9	Open the manifold valves.
10	Switch on F1 / F22.

1.9 Replacement of the primary cooling water hoses of the Helium compressor/COOLPAK



Increasing the water flow through the helium compressor, the old hose set must be replaced by a new water hose set. The old hose set includes a flow limiter and quick connects. The new hose set is necessary for Sonata systems with a new type of water cooled system (Part number 7121911 KE701)

1.9.1 Replacement kit and tools

Fig. 31: Water hose kit



- Two hoses with Armaflex insulation, with ball valves at one end.
- Two 90 degree elbows.
- Adhesive insulation tape and additional pieces of Armaflex insulation.
- Two hose clamps.

Tools:

- Bucket and towel
- Open wrenches 5mm, 19mm and 22mm or adjustable wrench
- cable ties

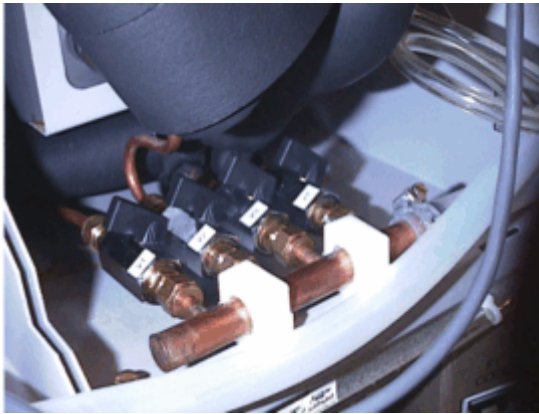
1.9.2 Prerequisites

- The MR system and the water cooling system (RCA) is switched off.
- The coldhead and the helium compressor is switched off.

1.9.3 Procedure of replacement

- Close the connection to the on-site primary water supply (water in/out); e.g. the two shut-off valves at the primary cooling water transfer panel, if installed.

Fig. 32: Drain valves in the RCA



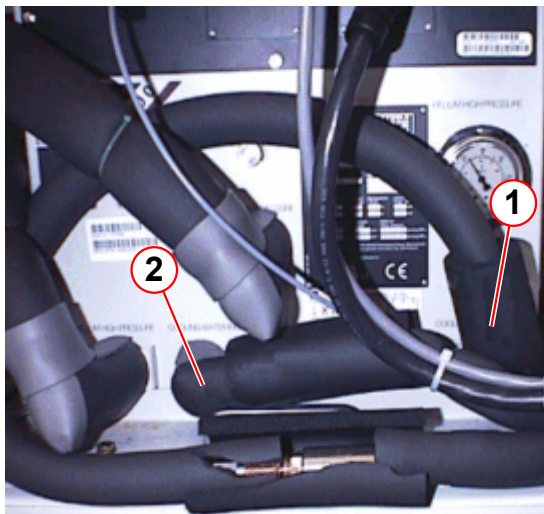
- Drain the primary water circuit by opening valves V3 and V4 (below cooler 1 in the RCA).

Fig. 33: RCA - right side cabinet wall removed



- Remove the right side panel of the RCA cabinet, to get access to the primary cooling water hoses to the coolpak.

Fig. 34: Water hose connections at the Coolpak



- (1) Water out
- (2) Water in

- Remove the Armaflex insulation of the water hoses at the helium compressor side, to get access to the quick couplings: Water in: Quick coupling at the 90 degree elbow
Water out: Quick coupling at the flow limiter (→ Fig. 35 Page 31).

Fig. 35: Water flow limiter



- Disconnect the quick coupling at the flow limiter (water return).

Fig. 36: Draining the hoses



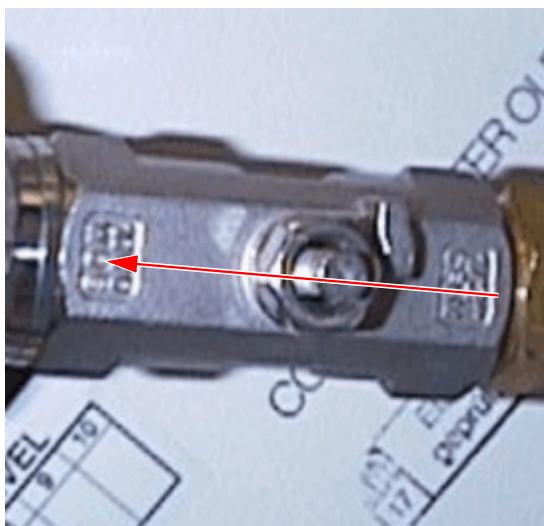
- Prepare a towel to wipe off spilled water.
- Drain the hoses into a bucket by pressing a screwdriver in the quick coupling at each side one after another.

Fig. 37: Removing water hose connections at manifold



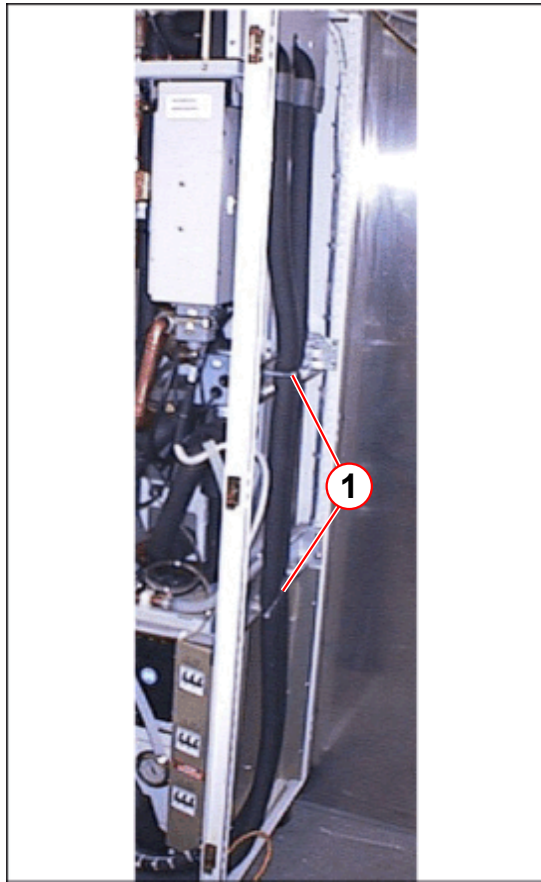
- Use 19 mm and 22 mm open end wrenches to remove the hoses from the adapters at the manifold in the RCA. Remove the hose connection with the 19mm wrench turning ccw, while holding the opposite with a 22 mm spanner. Hold a bucket ready to collect possibly draining water.
- Remove the hose connections (elbows) at the helium compressor.
- Remove the old hoses from the RCA.

Fig. 38: Open ball valve



- Check that the ball valves at the end of the new hoses are open; you can turn the ball valve with a 5 mm open end wrench.

Fig. 39: Routing water hoses



(1) Cable ties

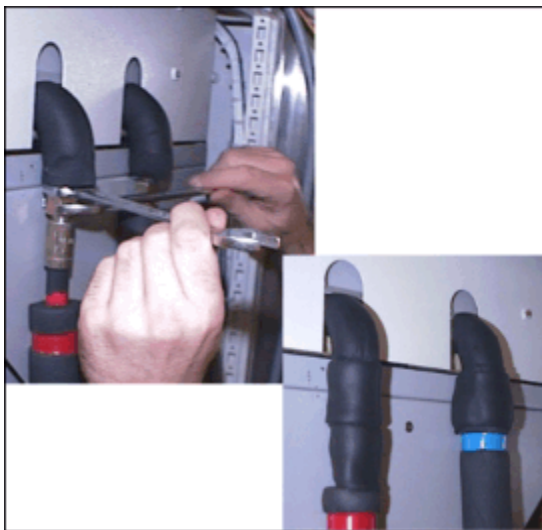
- Route the new water hoses in the RCA as shown in (→ Fig. 39 Page 33). Secure the hoses with cable ties at the cabinet frame (like the old ones).
Blue ring= inlet side, red ring= outlet side.
The 90 degree elbows must be connected to the helium compressor.

Fig. 40: Routing of water hoses



- Route the new hoses in front of the helium compressor as shown in (→ Fig. 40 Page 33).

Fig. 41: Connections at the manifold



- Connect the new water hoses at the manifold and at the helium compressor connections:
blue ring= inlet side, red ring= outlet side.
Screw on the hose connection with the 19 mm wrench turning cw, while holding the opposite with a 22 mm spanner.
- Insulate the connections with Armaflex adhesive tape as shown in (→ Fig. 41 Page 34), right side.

In case of an older type of manifold as shown in (→ Fig. 42 Page 34).

Fig. 42: Older type of hose nipples at the manifold



- Cut off the Kutting connections from the water hoses at the side of the manifold.
- Connect the water hoses with a hose clamp at the hose nipples of the manifold.

Fig. 43: Connections of water hoses at the Coolpak



- Mount the 90 degree elbows at the helium compressor.
- Connect the ball valves with the elbow as shown in (→ Fig. 43 Page 35).
Blue ring= inlet side, red ring= outlet side.
No extra gaskets are needed.
- Insulate the connections with Armaflex adhesive tape.

Note:

Provide for a good Armaflex insulation without gaps of the primary cooling water hoses and connections in the RCA to prevent water condensation.



The start-up of the primary cooling water circuit (i.e. refilling of cooler 2, hoses to the helium compressor and CCS) should be arranged with the responsible people at site.

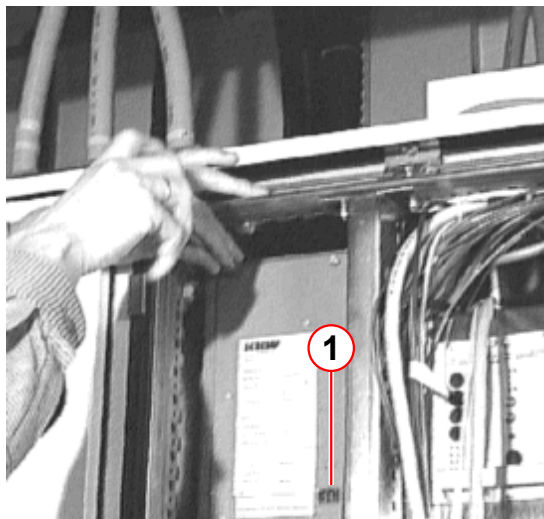


Before refilling the primary cooling water circuit, be sure that the drain valves V3 and V4 below cooler 1 in the RCA are closed.

1.10 Cabinet Cooling System

1.10.1 Replacing the ventilation unit

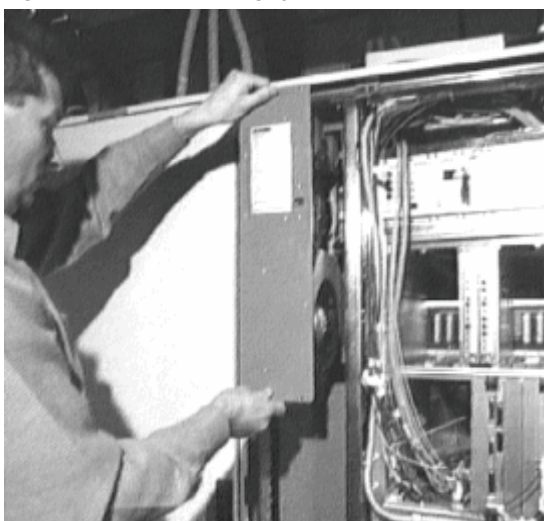
Fig. 44: Cabinet cooling system



(1) Connector X800

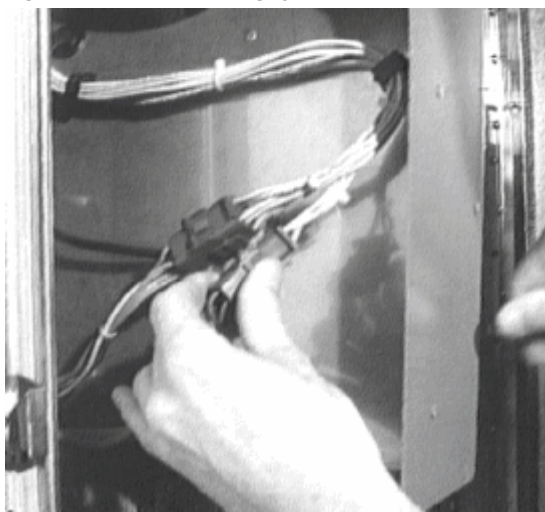
- Switch off F10.
- Disconnect cable X800.
- Remove the 4 screws Allen screw 4 mm.

Fig. 45: Cabinet cooling system



- Slide out the ventilation unit so that you can get to the cable from the side.

Fig. 46: Cabinet cooling system



- Disconnect cable X5/X6 and X3/X4.
- You can now pull out the ventilation unit.

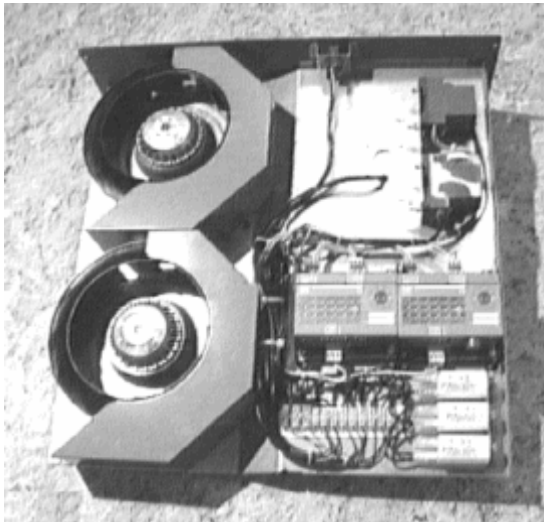
Fig. 47: Cabinet cooling system



- Replace the defective component.

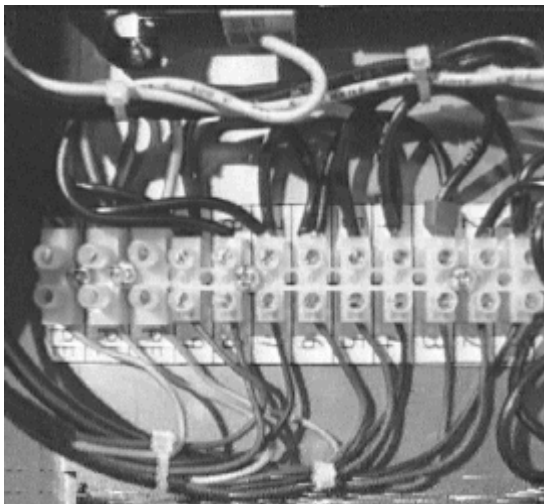
1.10.2 Replacing the blower

Fig. 48: Ventilation unit



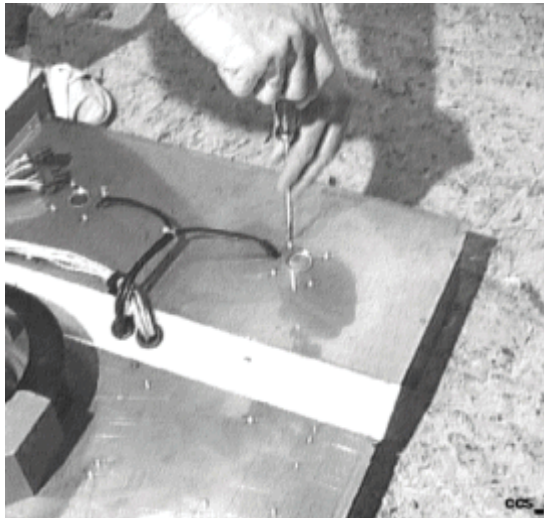
- Remove the ventilation unit.

Fig. 49: Ventilation unit



- Disconnect the cable of blower M1, M2 or M3.
M1: X2: 1(bl), 2(br), 3(bl); PE
M2: X2: 4, 5, 6; PE
M3: X2: 7, 8, 9; PE

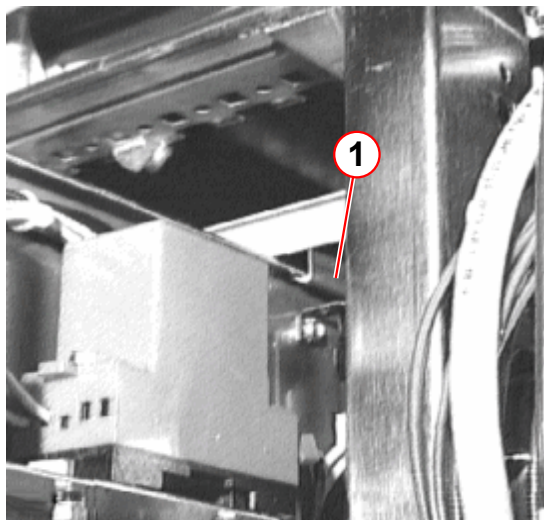
Fig. 50: Disassembly - Blower



- Loosen the screw of the blower.
- Install the new blower in the reverse order.
- Connect the cables on X2.

1.10.3 Installing the ventilation unit

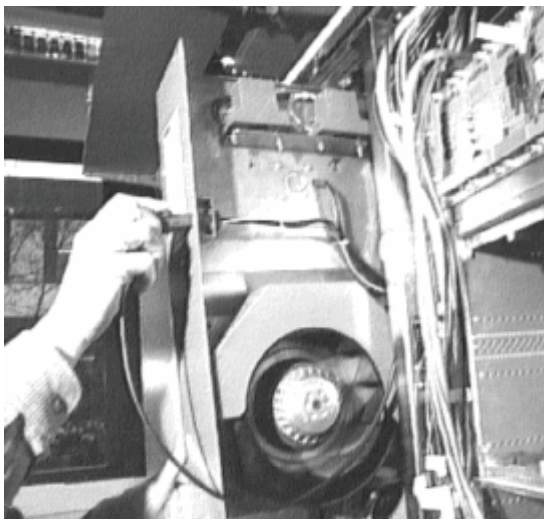
Fig. 51: Ventilation unit



(1) Guide rail

- Slide ventilation unit into the guide rail.

Fig. 52: Ventilation unit



Check the functions of the ventilation unit:

- Connect cable X5/X6 and cable X3/X4.
- Connect cable X800 and switch on F10.
- Slide ventilation unit back in and secure with screws.

1.11 Replacing the water-cooled CCS



If a new type of water-cooled CCS is installed, the primary water hose set to the helium compressor/ COOLPAK must be replaced.



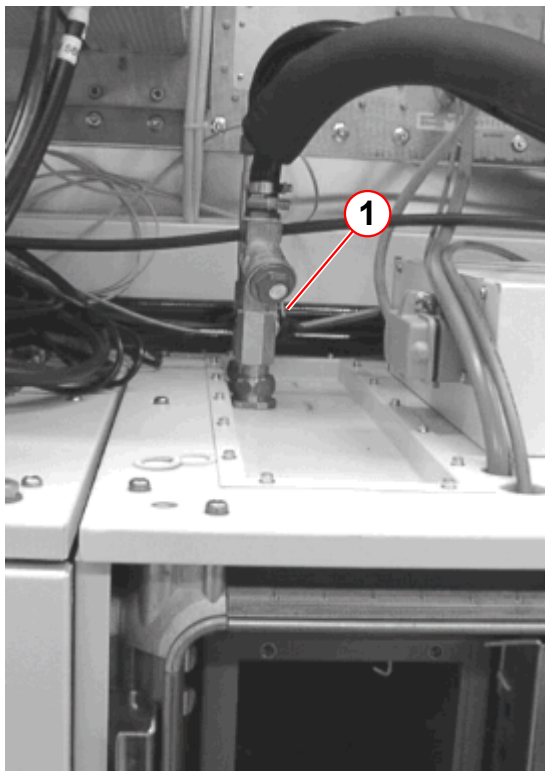
Parts with knurled threads has to be sealed with hemp or teflon tape.

1.11.1 Auxiliary tools

- Torque wrench 25 - 130 Nm (342456)
Open end insert (7548444)
- Spanner SW 36 mm;
- Socket wrench SW 41 mm
- Screwdrivers
- Adjustable spanner
- Pipe wrench, adjustable
- Allen key: 4 mm, 3 mm
- Bucket

1.11.2 Removing the heat exchanger

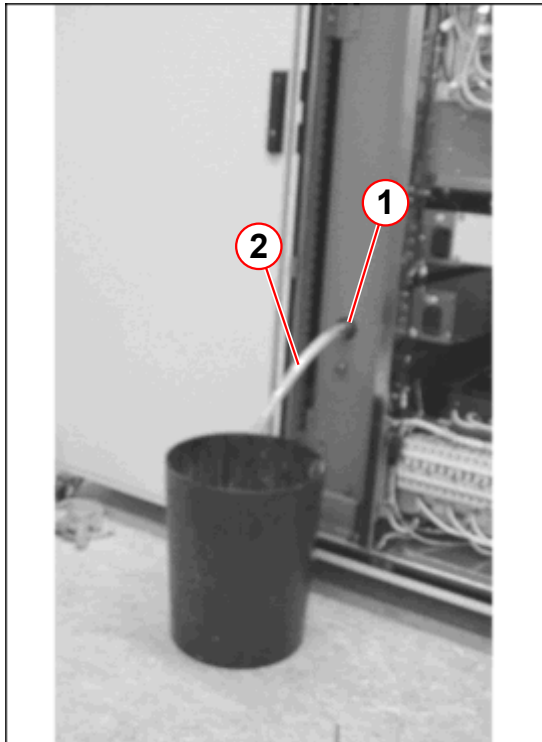
Fig. 53: Water-cooled CCS



(1) Shut-off valves

- Switch off chilled water supply and return, close the valves.

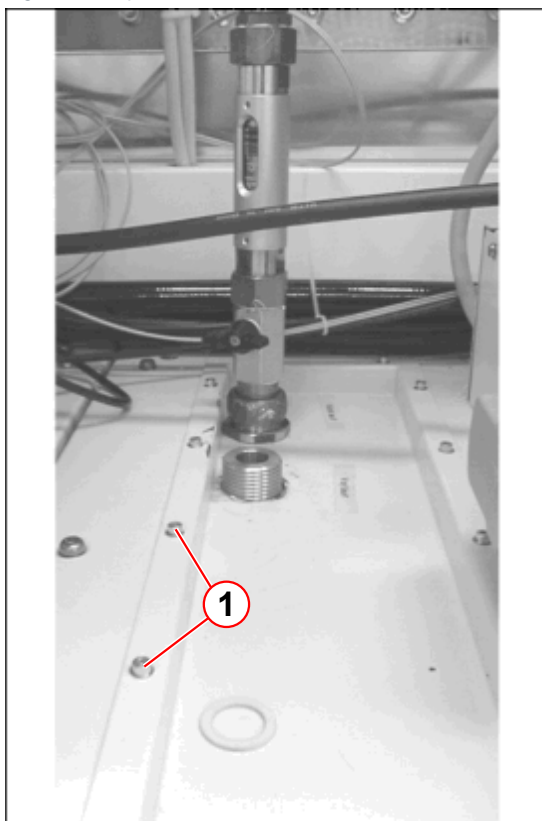
Fig. 54: Water-cooled CCS



- (1) Drain valve
- (2) Drain hose

- Drain CCS, open the drain valve.
CCS must be on power for opening the inlet valve during draining.
- Vent CCS. Loosen the screws at the connection on top of the cabinet.

Fig. 55: Top of the RCA



(1) Screws for the top cover

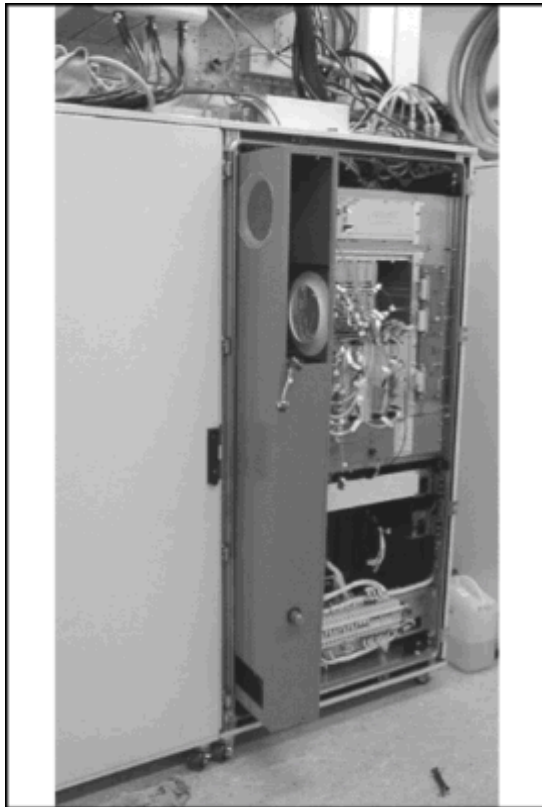
- Remove flow and return.
Open-end wrench SW 36

Fig. 56: Top of the RCA



- Take off top cover
Allen key 3 mm
For water-cooled CCS only!
- Take off the insulation parts
- Remove ventilation unit.
For reducing the weight of the heat exchanger

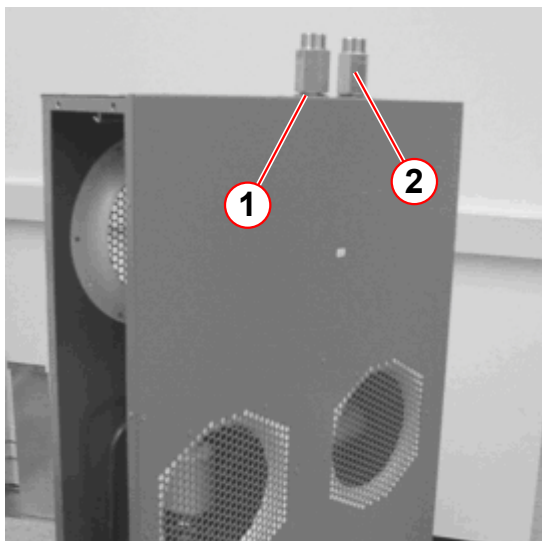
Fig. 57: Cabinet cooling system



- Remove CCS mounting bracket.
- Remove the condensate pump (if it exists).
- Slide heat exchanger out (towards you) .
You can get to the CCS from the side by swivelling out the MMC rack.
- Remove hose connections beforehand, if necessary Open-end wrench SW 41.

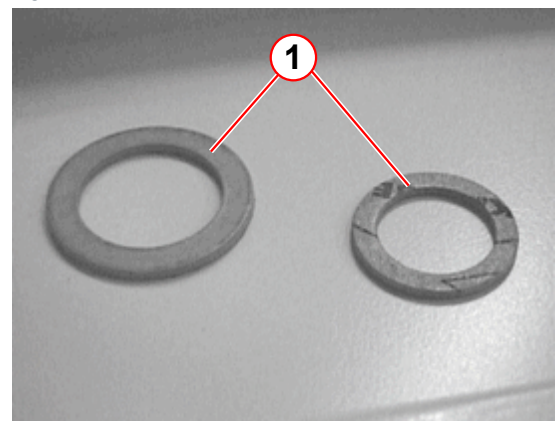
1.11.3 Installing the heat exchanger

Fig. 58: CCS after it has been removed



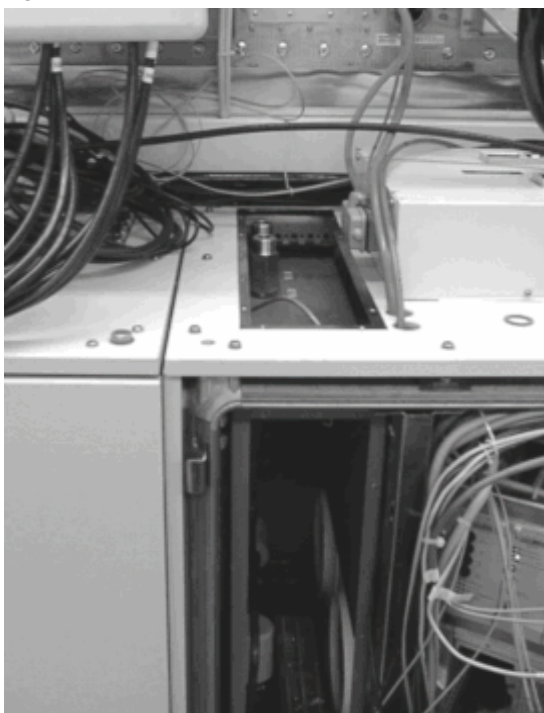
- (1) Washer
(2) Hose connections

Fig. 59: Washers



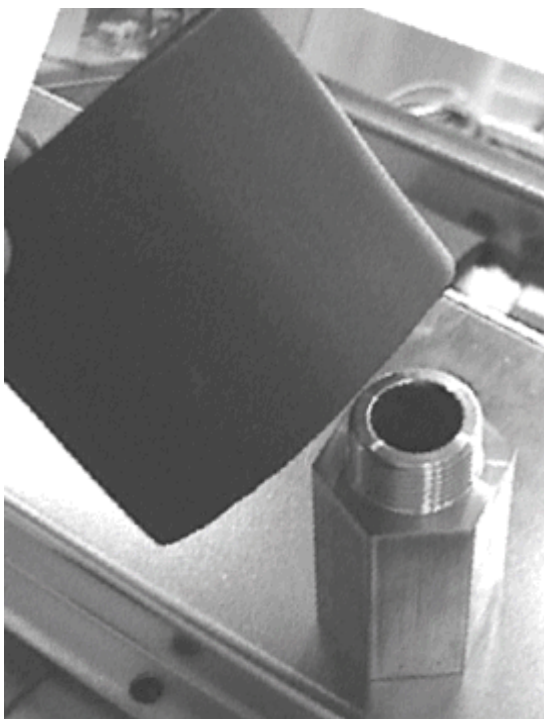
- (1) Washers

Fig. 60: Water connections



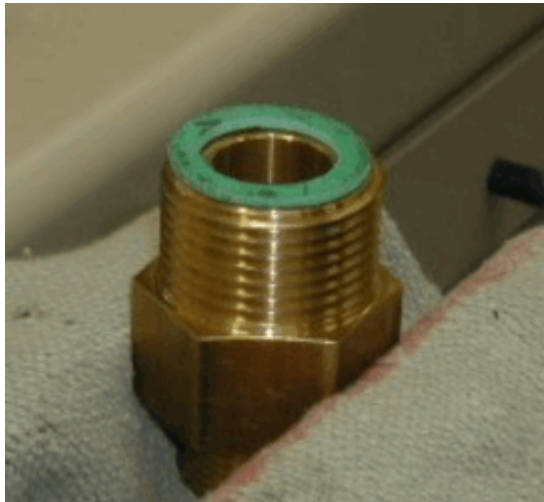
- Slide heat exchanger into the cabinet.
- Insert hose connections including new washers and secure them .
Socket wrench SW 41.

Fig. 61: Water connections



- Insulate the hose connections.
- Mount the top cover.

Fig. 62: Gasket



- Use the gaskets as shown.



To avoid problems due to cracked joints, all of the flat-sealing screw joints must be torqued to 50 Nm.

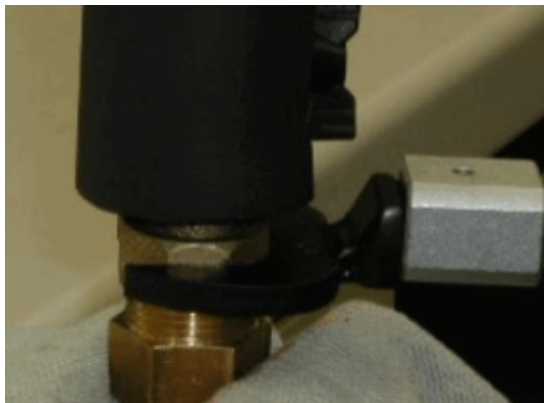
Use the special open end inserts (7548444) for the torque wrench 25 - 130 Nm (3424561).

Fig. 63: Torque wrench



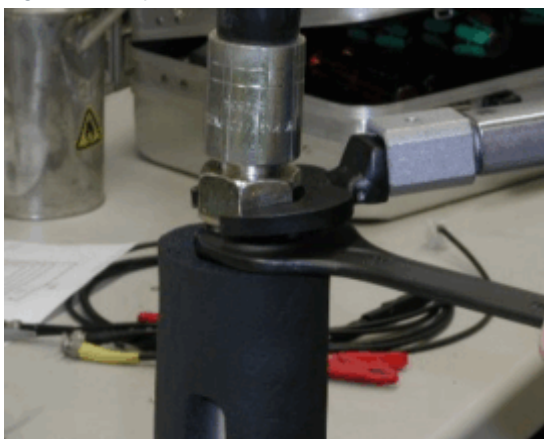
- Set the torque wrench to 50 Nm

Fig. 64: Torque wrench



- Use the torque wrench as shown here to tighten the flat-sealing screw joints of the flow meter and the filter element.

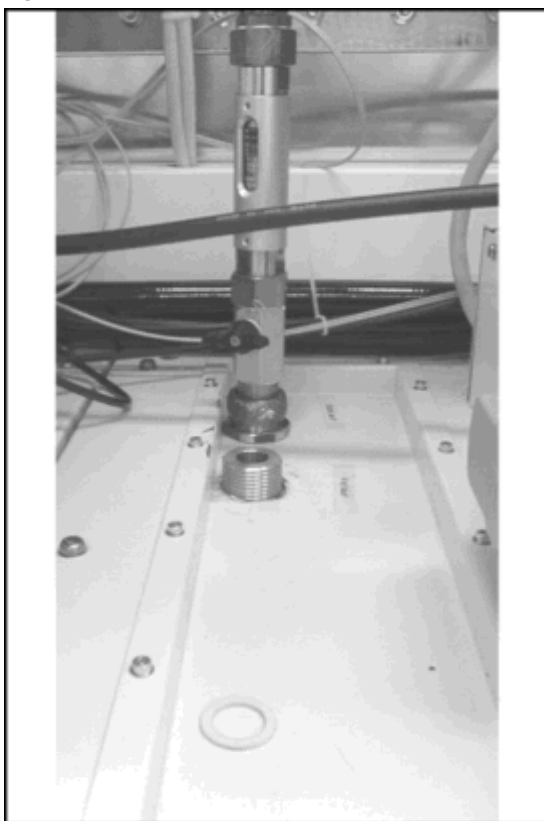
Fig. 65: Torque wrench



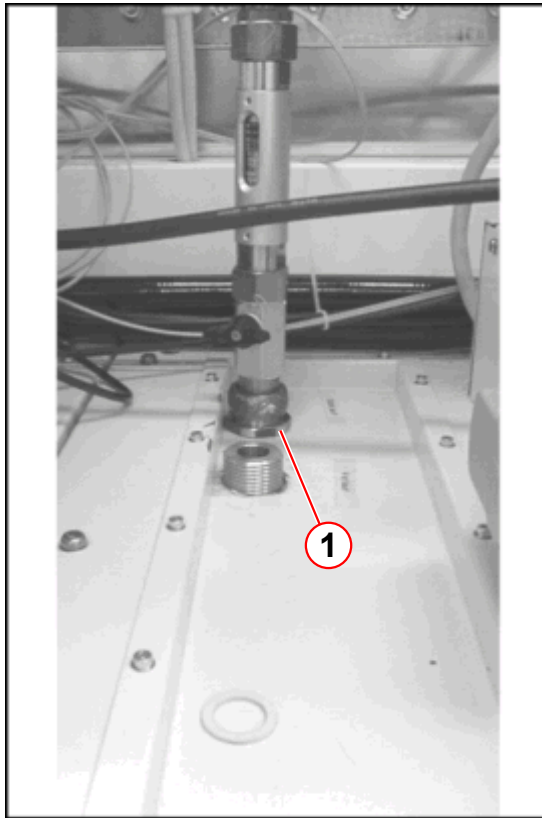
- Use the torque wrench as shown in this example to attach the hoses to the top of the flow meter and the filter element.

Always use a second spanner to prevent the flow meter or the filter element from turning.

Fig. 66: Water connections

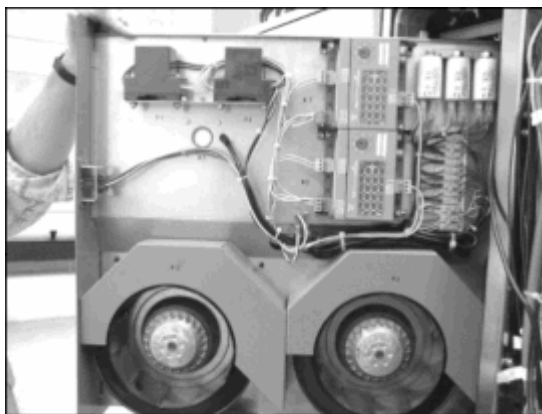


- Connect flow and supply.

Fig. 67: Ventilation unit

(1) Washer

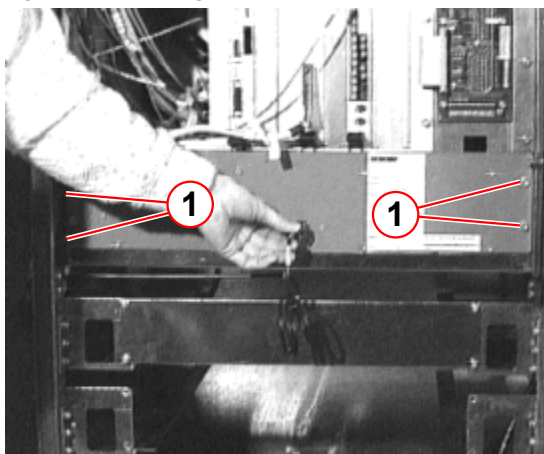
- Attach CCS mounting bracket.
- Verify that CCS drain valve is closed.
- Open shut-off valves.
- Check connections for leaks.

Fig. 68: Ventilation unit

- Install ventilation unit.
- Connect cable X5/X6; X3/X4.
- Secure ventilation unit.
- Connect cable X800.
- Switch on F10.
- Switch on MR system so that the flow valve opens.
- Check connections for leaks.

1.11.4 Removing the diffuser

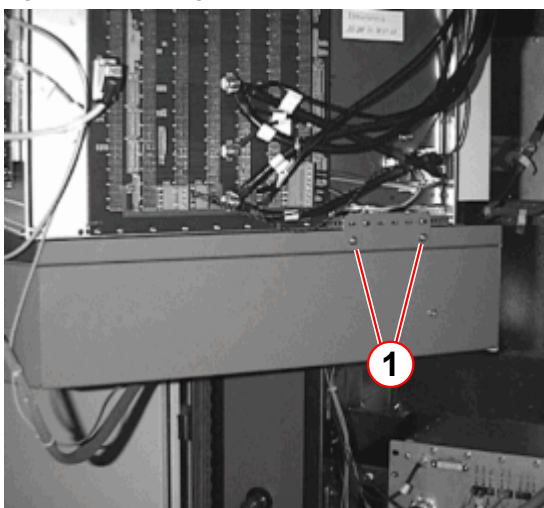
Fig. 69: Removing the air diffuser



(1) Mounting screws

- Switch off F10.
- Pull connector X801.

Fig. 70: Removing the air diffuser



(1) Mounting screws

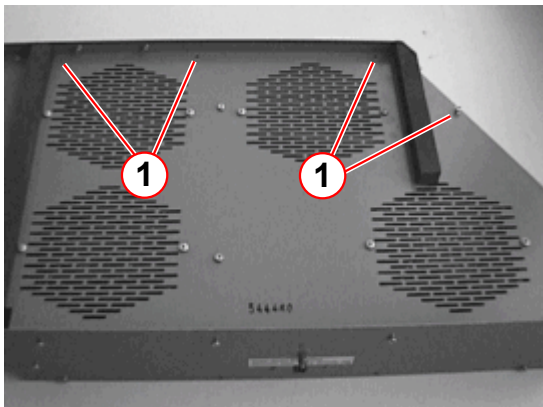
- Remove the 4 mounting screws Allen key: 5 mm
- Swivel out the MMC rack.
- Remove the 2 mounting screws in back.

Fig. 71: Removing the air diffuser



- Slide air diffuser forward (towards you).

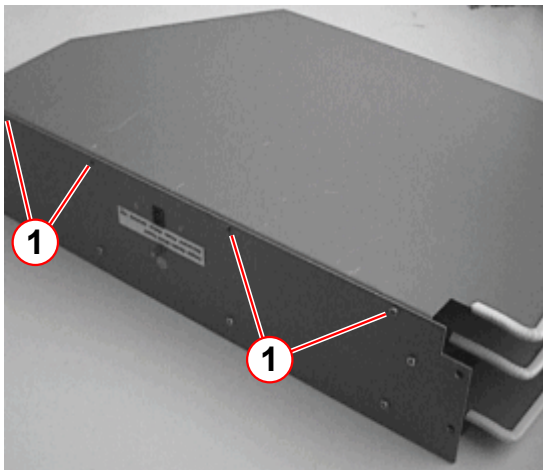
Fig. 72: Air diffuser



- Remove only the 4 screws as shown in (→ Fig. 72 Page 51)

(1) Screws

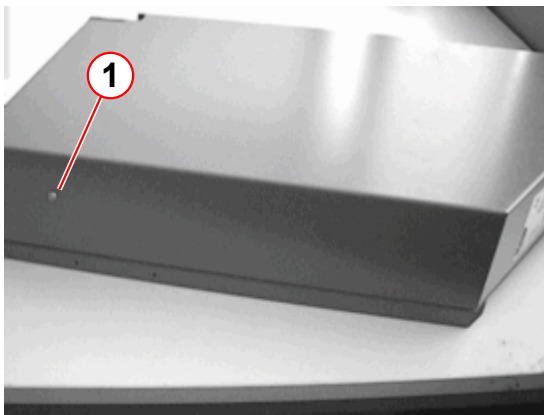
Fig. 73: Air diffuser



- Turn air diffuser by 180°.
- Remove the 4 screws in the front plate.

(1) Screws

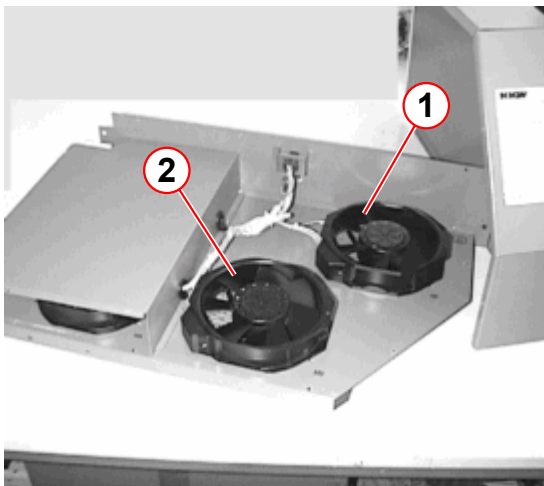
Fig. 74: Air diffuser



(1) Screw

- Remove one screw in the back.

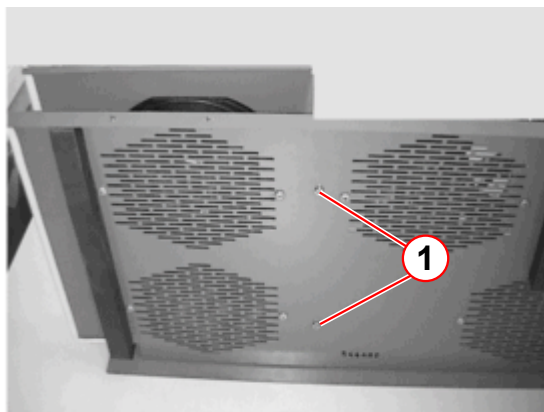
Fig. 75: Intermediate plate



(1) M1
(2) M2

- Take off cover.
- You can now remove M1 and M2.

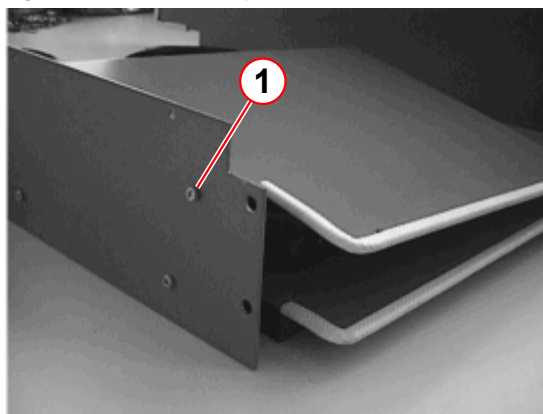
Fig. 76: Intermediate plate



(1) Screws

- Remove 2 screws.

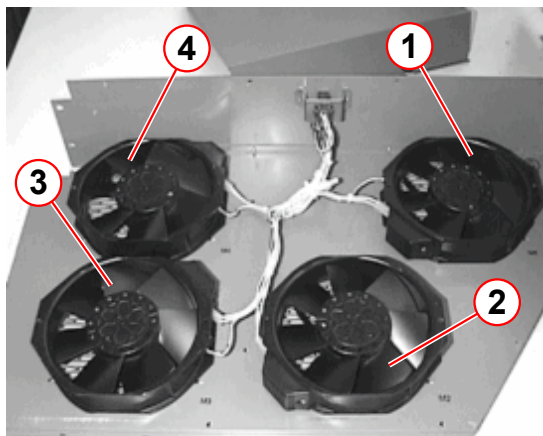
Fig. 77: Intermediate plate



(1) Screw

- Remove screw.
- Remove intermediate plate.

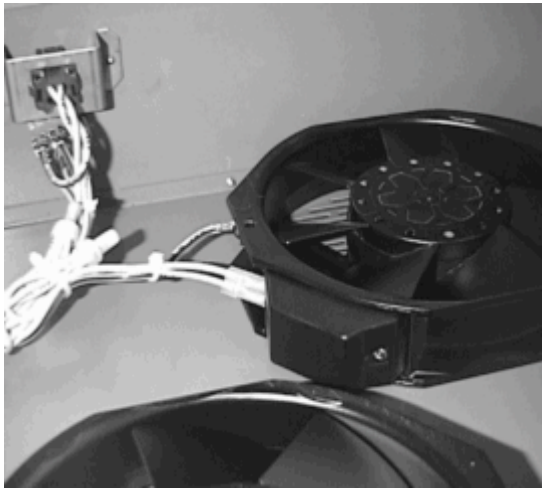
Fig. 78: Removing the ventilators



(1) M1
(2) M2
(3) M3
(4) M4

- You can access the ventilators.

Fig. 79: Removing the ventilators



- Take off the screws to remove ventilators.
- Disconnect cable.
- Disconnect protective conductor.

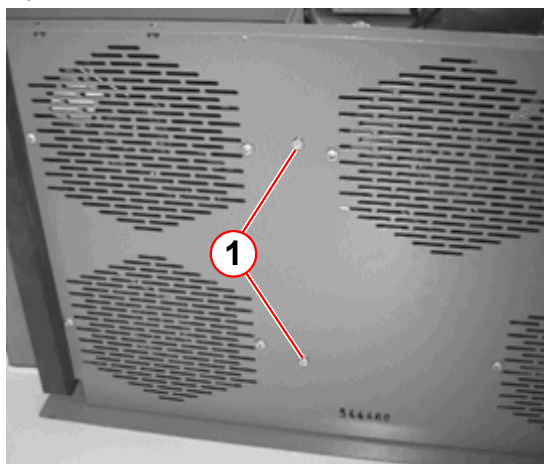
1.11.5 Assembling and installing the air diffuser

Fig. 80: Air diffuser



- Position intermediate plate.
- Route cable through the conduit.
- Bundle the cables so that they do not collide with the ventilator blades.

Fig. 81: Air diffuser



(1) Screws

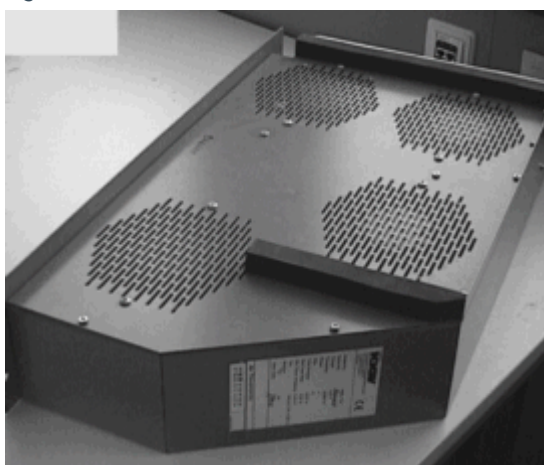
- Secure intermediate plate.
- Secure intermediate plate to back panel.

Fig. 82: Air diffuser



- Position cover and secure it.

Fig. 83: Air diffuser



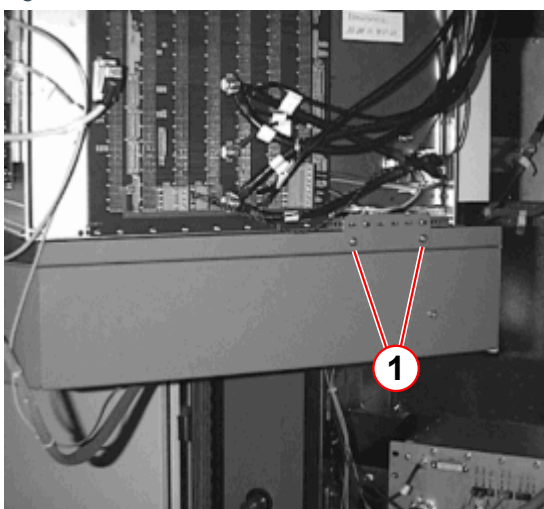
- Secure cover from the other side.

Fig. 84: Air diffuser



- Install air diffuser.

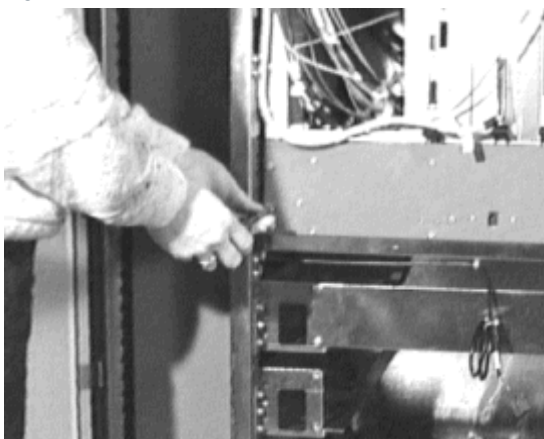
Fig. 85: Air diffuser



- Attach air diffuser in back.

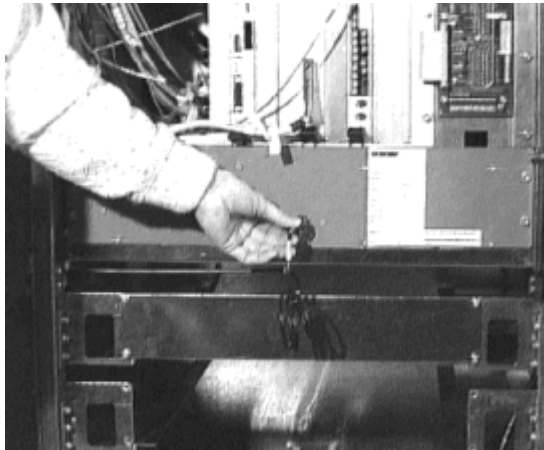
(1) Mounting screws

Fig. 86: Air diffuser



- Secure air diffuser in front.

Fig. 87: Air diffusor



- Connect cable X801.
- Switch on F10.

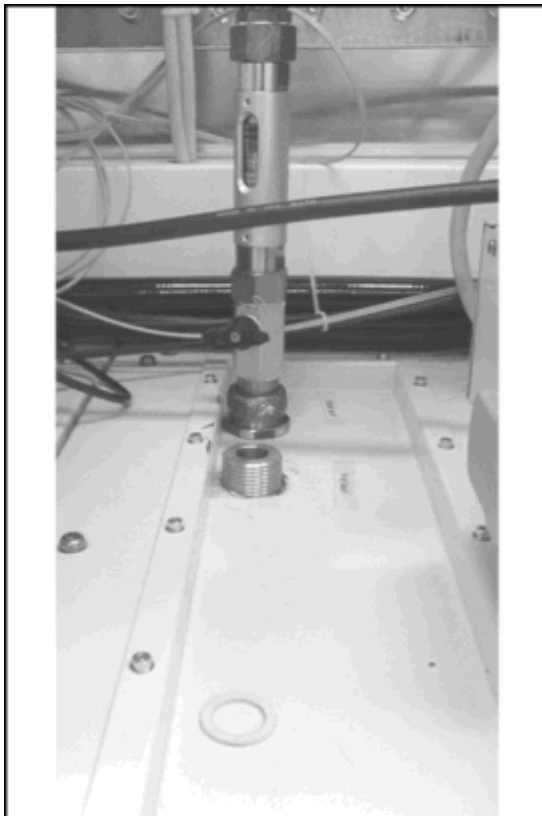
1.12 Replacing the flow meter



To avoid problems due to cracked joints, all of the flat-sealing screw joints must be torqued to 50 Nm.

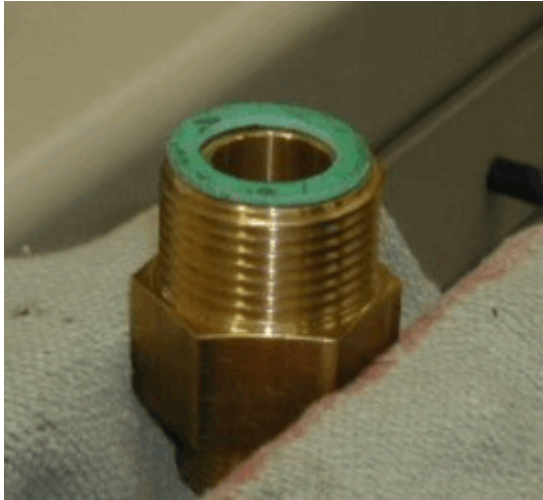
Use the special open end inserts (7548444) for the torque wrench 25 - 130 Nm (3424561).

Fig. 88: Flow meter



(1) Washer

Fig. 89: Gasket



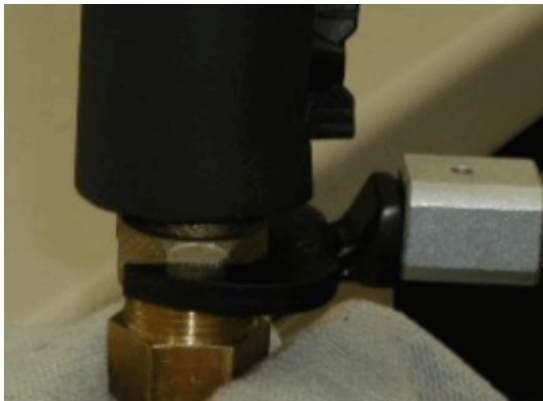
- Use the gaskets as shown.

Fig. 90: Torque wrench



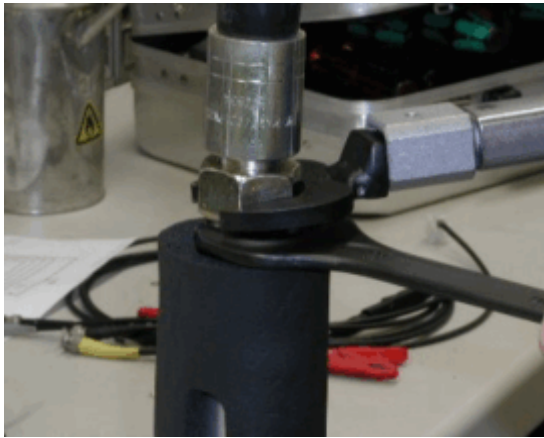
- Set the torque wrench to 50 Nm.

Fig. 91: Torque wrench



- Use the torque wrench as shown here to tighten the flat-sealing screw joints of the flow meter and the filter element.

Fig. 92: Torque wrench



- Use the torque wrench as shown in this example to attach the hoses to the top of the flow meter and the filter element.

Always use a second spanner to prevent the flow meter or the filter element from turning.

1.13 Drain procedure of the cooler

The following instruction cover the procedure to ensure that the cooler is completely empty of water.

NOTICE

Cooling water containing citric acid should be collected while draining the Cooler.

» This water should be refilled into the secondary cooling circuit.



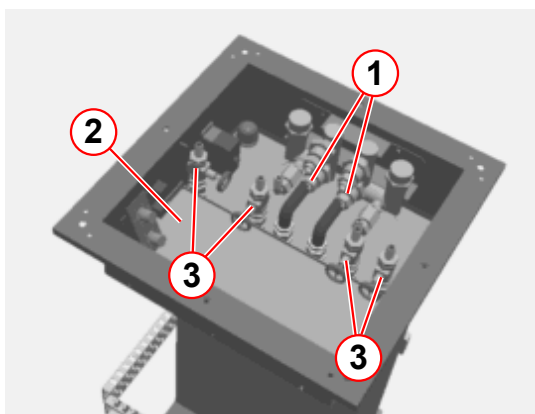
Follow the country-specific guidelines when disposing of cooling water containing citric acid.

1.13.1 Required Tools

- Spanner: 30 mm
- Screwdrivers
- Adjustable spanner
- Pipe wrench, adjustable
- Bucket

1.13.2 Procedure

Fig. 93: Top of RCA without water pipes

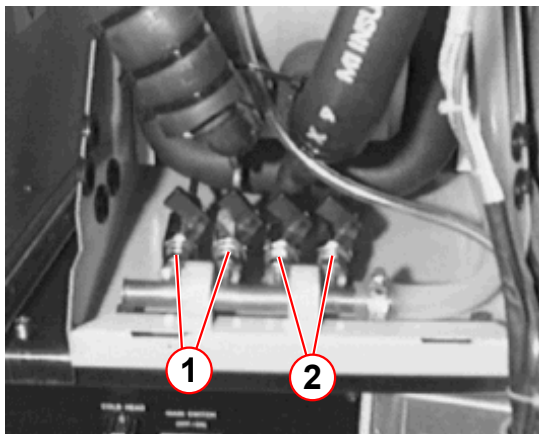


- (1) Primary water circuit
- (2) Top cover
- (3) Secondary water circuit

- Switch off voltage supply to the cooler.
Switch off circuit breaker F1 or F2 in the RCA
- Turn off primary water, supply and return.
- Turn off secondary water, gradient coil; electronic suit; flow and return

- Remove the front cover.
- Position the bucket in front of the cooler and put the cooler drain hose in it (the cooler holds approximately 30 l of water).

Fig. 94: Drain valves



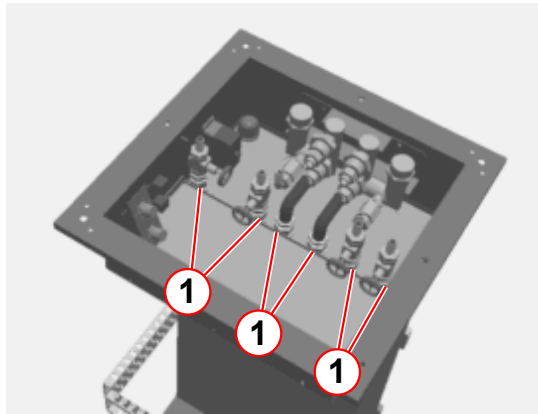
- Open all four drain valves
2 valves for primary
2 valves for secondary

Fig. 95: Bleeding valves



- Open the bleeder valves
The air entering the system will help drain the unit.

Fig. 96: Union end



() Union end

Cooler without bleeding valves

- Loosen the union ends on the primary "flow" and primary "return".
The air entering the system will help drain the unit.
- Loosen the union ends on the secondary "flow" and secondary "return".
This air entering the system will help drain the unit.

1.14 Filling the secondary water circuit

NOTICE

Fill water only when lamp LP1 lights up

Use distilled water only.

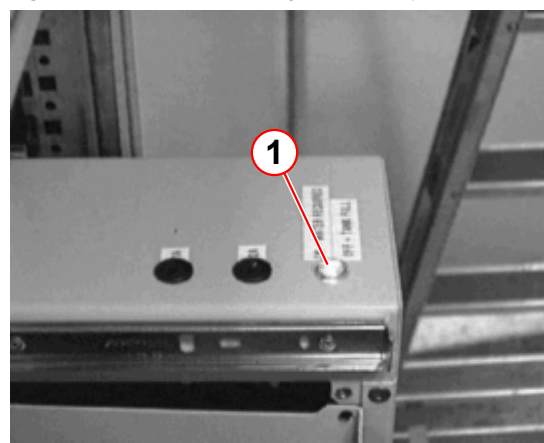
» Do not use any inhibitor.

- Switch on the cooler
 - Open the cap of the water tank and insert the funnel
 - Fill the water tank with distilled water
- Monitor the Yellow lamp, refer (→ Fig. 98 Page 64)

Fig. 97: Filling procedure at RCA cabinet



Fig. 98: Water indicator - yellow Lamp LP1



- Refill the water tank, until the yellow lamp goes out.

NOTICE

Stop filling immediately after lamp LP1 goes out.

» Avoid overfilling!

- » The common print number (MR-000.841.01) from Replacement of Parts is divided into separate Component print numbers.

Version 02

- Hint: "Protective earth connection test" was added.

Version 03

- New type 8022 flow meter added (→ Fig. 25 Page 23)

Version 04

- New procedure added:(→ Drain procedure of the cooler / Page 61) and (→ Filling the secondary water circuit / Page 64)

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Print No.: MR-000.841.18.04.02 | Replaces: MR-000.841.18.03.02
Doc. Gen. Date: 06.15 | Language: English
Part No.: n.a.
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